

GHATS

USER'S MANUAL

Version 3.3.0 - 29 August 2025

General
High-Energy
Aperiodic
Timing
Software



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From 26 August 2023, and after the passing of Tomaso), development has been taken over by:

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NOISE, n. A stench in the ear. Undomesticated music. The chief product and authenticating sign of civilization.

(A. Bierce “The Devil’s Dictionary”)

It is of a flawless white color, right from its nose to its magnificent tail.

(Mahabharata, The Story of Garuda)

Revision history

VERSION 1.0.1

Fixed uppercase/lowercase for *POWER/FFT* in *gh_xte*
Fixed crash in *GH_CROSS*
Added *GH_PLOT_LAG*
Added *GH_PLOT_COH*
Fixed wrong stretch time in *GHATS_ALL* output
Fixed wrong stretch time in *GH_XTE* output to terminal and log file

VERSION 1.0.2

GH_CROSS_RANGE added
GH_PLOT_CROSS removed (duplicate of *GH_PLOT_LAG*)
GH_XTE Fixed bug for sliding in turbo mode
GH_XTE Fixed bug for recalculation of number of PDS in non-turbo
GH_XTE Fixed bug: now it finds housekeeping files in turbo mode even if data are elsewhere
GH_XTE Removed calls to *FSC_FILE_BASENAME*, also removed from package for compatibility
GH_XTE Channel range now restricted to available, not claimed to be the same in input
GH_CROSS_RANGE Intercept case when zero or one bins are to be averaged
GH_CROSS/GH_CROSS_RANGE /TIMELAG keyword changed to TLAG for uniqueness
GH_CROSS/GH_CROSS_RANGE Fixed bug for time selection
GH_DYNA Changed default frequency rebinning from -100 to 1 (no rebinning)
GH_COLORS added
GHX/GH_DYN/GH_COLORS/GH_HK/GH_CROSS/GH_CROSS_RANGE fixed bug for times

VERSION 1.1.0

GH_XTE Added windowing option
GH_SWIFT added
GH_HK renamed *GH_HK_XTE*
GH_PLOT_HK renamed *GH_PLOT_HK_XTE*
GH_XTE Increased max number of files to 1000
GH_XTE/GH_SWIFT fixed but for empty data stretch
GH_CROSS Poissonian estimate was valid only for RXTE

Revision history (cont'd)

VERSION 1.1.1

GH_CROSS_RANGE Added output of average cross-spectrum value (complex)
GH_XMM Added
GH_NUSTAR Added

VERSION 2.0.0

GH_CROSS_RANGE Fixed bug for time lag output
GH_XTE Fixed bug that limited time rebinning
GH_SWIFT Fixed bug that limited time rebinning
GH_XMM Fixed bug that limited time rebinding
GH_NUSTAR Fixed bug that limited time rebinning
GH_CROSS Fixed for release of logical units
GH_CROSS Fixed bug for Poissonian estimate for instruments different from PCA
GH_CROSS_RANGE Fixed for release of logical units
GH_NUSTAR Increased maximum number of GTI intervals
GH_PLOT_HK_XTE Bug fixed
GHX Bug with file logical unit fixed
GH_XTE/SWIFT/XMM/NUSTAR Bug with input from command line fixed
GH_LAXPC Added
GH_CZTI Added
GH_CROSS_RANGE Fixed bug for time lag computation
GHX Indices for SEL option now start from 0 (IDL WHERE compatibility)

VERSION 3.2.0

GH_CROSS Added RATE keyword
GH_CROSS absolute value check for time difference for file compatibility
GH_CROSS added /poivalue keyword for real part of Poissonian contribution to be subtracted
GH_CROSS added keyword for cross-spectrum output
*GH_** added reading of TIMEZERO keyword from GTI files and extensions

Revision history (cont'd)

VERSION 3.3.0

First release after the passing of Tomaso Belloni

GH_CROSS_RE_IM_NEW Added

GH_CROSS_RE_IM_NEW Computes de Real/Imaginary parts of the cross spectrum

GH_CROSS_RE_IM_NEW Can rotate the cross spectrum by 45 degrees

GH_CS Added

GH_CS Uses *GH_CROSS_RE_IM_NEW* twice; in the process, it corrects for deadtime induced cross talk and, if necessary, for partial correlation if the subject and reference bands overlap

GH_CS Added the option to compute errors of the coherence in the case of high powers and low coherence (eq. 9 in Vaughan & Nowak 1997) using *GH_LOWCOH_SSQ* and *GH_LOWCOH_SSQ_CI*

CONCATENATE_PDS Added

CONCATENATE_PDS Concatenates PDS files

CONCATENATE_FFT Added

CONCATENATE_FFT Concatenates FFT files

GH_DYN_FITS Saves and plots a time-frequency image (a.k.a. Spectrogram) from *GH_DYN* to a *FITS* file

GH_DYN_FITS_TK.py *GUI-python* interactive tool to plot spectrograms created with *GH_DYN_FITS*

Introduction

(This was the original Introduction, written by Tomaso Belloni)

The need for a timing package

The long-living RossiXTE mission was aimed at observing time variable phenomena from X-ray sources. Its main instrument, a large proportional counter which, in addition to its large collecting area, provided high time resolution with good calibration and knowledge of dead time effects, was designed for the study of fast variability. While the XRONOS package distributed by NASA/HEASARC can provide tools for the timing analysis of periodic phenomena such as pulsars, unfortunately, no comprehensive and efficient tool was provided to the community for the aperiodic timing analysis of RXTE data and none existed before. Each group was left to their own devices and had to develop a custom package.

The background behind GHATS

At INAF-OAB, more than ten years ago, I started writing a package devoted to RXTE/PCA data (the HEXTE instrument, although called High-Energy X-ray Timing Experiment, is almost unusable when it comes to timing). The philosophy behind the software can be summarized in two points:

1. The program had to work directly from RXTE raw files, without forcing the user to produce light curves before the analysis. In other words, the program itself would accumulate its own light curves, which would be discarded at the end.

2. Timing analysis based on Fourier Transforms is essentially based upon two separate steps. The first is the production of Power Density Spectra (PDS) or Fast Fourier Transforms (FFT). The second is the analysis of the products. The first task is essentially an almost non-interactive process: one gives input parameters and the program delivers the final product (there is some interaction as the range of possible parameters must depend on the specific data types and a decision can only be made after reading the input files). The second task is highly interactive: read PDS files, manipulate them, fit them.

Given these starting points, the best solution was to sacrifice some speed and write the software in an integrated (graphical) environment. Although yours truly is not a fan of IDL, which I find extremely cumbersome and incredibly old-fashioned to the point of reflecting what nowadays can only be defined as bad programming, the choice fell upon IDL. The reasons are simple: it's a very widely used package and there are existing libraries which make life easier. The downside of it is that IDL is not a freely available program and must be bought from a commercial source, which effectively limits its availability. However, since years considerable effort is going into the development of the GNU Data Language (GDL), essentially a public-domain clone of IDL. Although it is by far not complete, GDL is at a sufficiently advanced stage to be used for my purposes.

A new package with limited distribution

From all this, a package called MU was born. Initially only used within my institute, it was moderately exported so that it is now used by very selected people in Groningen, Rome, Mumbai and Kolkata. It worked only on RXTE/PCA data and continues to do so. The upcoming ASTROSAT mission provided the incentive to extend it and adapt it to other instruments. Although the package was in a good shape, the data format was too RXTE specific and needed to be changed and made more general.

Introduction

From the old to the new

This is how GHATS was born. Release 1.0.0 had not been used by anyone other than me for testing. Then the package was released. The few users of MU will recognize essentially the same software, with only command names changed. This is because the changes in data format are of course invisible and the extension to other missions are additions to the existing code. In due time, specific extensions to ASTROSAT will be available.

The future

What GHATS needs now is testing by a larger number of users. In particular, the GDL version of MU has been used only by one person until now and certainly will profit from more users.

New Introduction

(This New Introduction was written after the passing of Tomaso Belloni, and for the release of GHATS 3.3.0)

A new start

We, Sara, Federico and Mariano, have used GHATS for several years, had discussed many times with Tomaso about ways to improve it, bugs, etc., and have contributed parts of some of the routines.

The last version of the code that we received officially from Tomaso was 3.2.0, from February 2023, about 6 months before he passed away.

Since then, we in Groningen made a few small changes to the existing routines, and expanded the code with a number of new routines that allowed us to study the cross spectrum and the coherence function.

As a legacy to his work, we decided to upload Tomaso's last version, 3.2.0, to this Github, and start a new version, 3.3.0, that includes those new routines. This is the version that will now be developed further.

Besides adding those routines, we updated the Manual (we left the old version intact in 3.2.0) and the links to the necessary libraries, included instructions to install and run GHATS in GDL, incorporated a PYTHON directory to add new Python-based scripts for GHATS, and added a new Walkthrough for the cross spectrum.

We miss Tomaso, and his enthusiasm in adding stuff to the code whenever we came with ideas. We would have very much liked to share this new step with him. In the meantime, we try to honour his memory by trying to do good work using this software. We are sure that you, the new generation of users, will do the same.

Mariano, Federico & Sara
Groningen, La Plata, Merate
November 2025

Installation

The package

Originally, the GHATS package could be obtained from INAF-OAB and was not distributed in any way. This was so because Tomaso did not have the resources to support the software for a broad community.

From 3.3.0 we decided to make the code available on this Github repository to the community. We, however, cannot support all the request for updates, upgrades, etc., and will allow the community to add contributed routines or modify parts of the code (e.g., to correct bugs) on a development branch.

Unpacking

You can unpack the package in your preferred directory. Notice that it contains only ASCII files, as the IDL/GDL distribution is through source code (GDL does not yet allow binary distribution). The tar file will unpack two files and four directories:

- FFT: directory with the files for FFT production
- ANALYSIS: directory for the interactive analysis
- CROSS_ANALYSIS: directory dedicated to the cross-spectrum analysis.
- PYTHON: *Python*-based GHATS tools. Must be added to your PATH variable

(see below)

- ghats: the executable for IDL
- gghats: the executable for GDL

plus a number of directories specific for the missions that are supported.

Move the executable you are going to use to a directory in your path and make it executable if it is not already so.

Software dependencies

You need to have the following libraries installed on your computer:

1. An updated and running version of IDL or GDL. Notice that if you find a precompiled GDL for your architecture, you still need the source files as many procedures are external. GDL can be installed using the script *scripts/build_gdl.sh* available in the Wiki guide (<https://github.com/gnudatalanguage/gdl/wiki/GDL-on-Linux>), following the compilation instructions.

2. An updated version of the IDL ASTRON library (from <https://asd.gsfc.nasa.gov/archive/idlastro/>).

3. An updated version of Craig Markwardt's IDL library (from <https://pages.physics.wisc.edu/~craigm/idl/idl.html>).

Not everything is needed, but it is mostly routines which are useful to have handy.

Local set up

To have GHATS running, you need to customize your ghats/gghats startup file. The standard version that comes with the package assumes you are bash or another Bourne-like shell (if you are using [t]cshell you will need to use setenv), and contains:

1) export MU_PATH=/Users/belloni/Desktop/GHATS/

Replace with the directory containing your GHATS installation.

2) export IDL_STARTUP=\$MU_PATH/ANALYSIS/ghats.pro

3) export IDL_DIR=/Applications/itt/idl/idl80

Replace with the location of your IDL/GDL directory

Installation

4) export IDL_PATH=\+\$IDL_DIR/lib:\+\$MU_PATH:\+/Users/belloni/gdl/src

Here you should export the locations of the IDL (or GDL) library. If you installed the additional libraries in different locations, add those as well.

5) export PATH=\$PATH:\$IDL_DIR/bin/idl

The path of the IDL/GDL executable, which is run from here.

6) export PATH=\$PATH:\$MU_PATH/PYTHON/

Add the PYTHON directory to your path.

Running the program

You then invoke GHATS running the appropriate script, for IDL or GDL, the commands are, respectively:

```
$ ghats
```

or

```
$ gghats
```

Below we give a couple of examples of the usage of the program (Walkthroughs).

Walkthrough

The data

The starting data are directly products from the instruments. In the case of RXTE data, this means science array and event data (SA and SE). Supported data modes are:

- Binned
- Single Bit
- Event
- GoodXenon (preprocessed)

Binned data contain photons within a range of channels from 0 up, Single Bit within a limited range, Event from a starting channel (which can be 0) up to the maximum. GoodXenon data are unbinned data, so they cover the full channel range. The time resolution and channel resolution of the data depends on the precise data type.

The input to the PDS/FFT production program is either a file, a list of files, or a list of lists of files.

- **Single file.** It must be one of the science PCA files (starting with FS37, FS3b, FS3f or FS4f).

- **List of files (a.k.a. metafile).** It is an ASCII file containing filenames (with pathnames if necessary), one per line. The files do not have to reside in the same directory or come from the same observation, but they must be homogeneous (same mode, same time resolution, same energy ranges and binning). They should not be simultaneous, but this is not usually possible. Time sorting is not needed. Metafiles are identified by a starting '@' character.

- **List of list of files.** It is an ASCII file containing a list of metafile names, one per line). The metafiles must themselves contain files which are homogeneous and do not overlap either in time or in channel space. The time resolutions do not have to be the same: the lowest one will be used by the program. Metametafiles are identified by a starting '@@'.

EXAMPLE: the observation contains the following files

Binned (B_2ms_4B_0_35_H)

FS37_473ed44-473fa98
FS37_4740390-4741118
FS37_4741b40-4742798

Event: (E_16us_16B_36_1s)

FS3b_473ed44-473fa9e
FS3b_4740390-474111e
FS3b_4741b40-474279e

Single bit: (SB_62us_0_35_500ms)

FS3f_473ed44-473fa9e
FS3f_47403a0-474111e
FS3f_4741b40-474279e

Walkthrough

Not usable: (D_1us_0_249_128_1s_F)

The data

The first group of files contains data binned into 2 ms time bins and four channel bins between channels 0 and 35 (a check on the RXTE Appendix E, ftp://legacy.gsfc.nasa.gov/xte/nra/appendix_f/RXTE_tech_append.pdf, will show that the channel boundaries are 0-13, 14-18, 19-25, 26-35).

The second group contains event (photon by photon) data with a time resolution of 16 microseconds and binned in 16 channels starting from channel 36. Therefore, binned and even data together cover the full channel range, but at a non-homogeneous time resolution. It is not possible to sample frequencies higher than 256 Hz below channel 35.

This limitation is solved by the third group of files, that contains data binned into 62 microsecond bins and only one channel between 0 and 35. We lose energy resolution but can go above the kHz.

What we can do is feed the program a single file (you pick), or a file called for instance @sb.lis containing a list of the single-bit files, or a file called @event.lis containing a list of the event files, or a file called @@all.lis containing @sb.lis on the first line and @event.lis on the second. To be specific:

```
$ ls FS3f* > @sb.lis
$ ls FS3b* > @event.lis
$ ls @sb.lis @event.lis > @@all.lis
```

The production program

Before running the software, unfortunately we cannot deal with compressed files. If the FS files above are gzipped, you need to unzip them. Also, you need to unzip a few auxiliary files which are going to be used by the software:

```
$ gzip -d FS4a* FH5a*
```

Start GHATS. We are now ready to start the production program, called **GH_XTE**. It can be called in three separate ways: without arguments, with arguments and with a parameter file. Let us first start with the interactive way, running it without any arguments

```
Ghats> gh_xte
```

If you are running IDL, you can use the **/TURBO** keyword, which switches to a different FFT engine (**IDL_EXTRACT** by C. Markwardt) and is faster.

A ‘file selection’ will appear, which by default will only show relevant files. In our case, let us choose @event.lis

We are shown the available channel ranges and asked whether we want all of them or a selection. Let us go for all (default).

Walkthrough

The production program

We are asked whether we want to produce a PDS, i.e. a file containing power spectra, or an FFT, a file containing Fourier transforms. Unless we plan to work on time lags or coherence function, a PDS is what we want. Let us choose POWER (the default).

Output file name. The default is the same name of the input file (without '@' in this case), with the extension PDS. Such extension is not mandatory, but it helps further defaults. Let us stick with event.pds.

We can rebin our data in time if we are not interested in the highest reachable frequencies. In our case the time resolution of the dataset is 16 microseconds. Since the RXTE clock works in powers of two, this means that we have 16384 time ticks per second. This means we can reach a Nyquist frequency of 8192 Hz, too high for our purposes. Let us rebin by a factor of 4 and go up only to 2048 Hz, sufficient to detect high-frequency features if present.

Next we have to tell the program how many time bins we want for each data stretch. The program will accumulate data until it has reached the required number of consecutive time bins, produce the PDS and store it. If a gap is reached before the number of bins is reached, it will discard the current interval and start again accumulation at the end of the gap. This will ensure that no gaps fall into data stretches. We have two ways to tell the program how long each data stretch should be. The first is to input the exact number of data points per stretch, which requires a calculation (division of the required length by the time resolution). The other is to input the length in time. Only powers of two are accepted in either case. The program discriminates between the two cases by checking whether you input an integer (number of bins) or real (time length) number. The program informs you beforehand how many points are needed for 16-second stretches (16 seconds is the accumulation time for RXTE/PCA spectra data). In our case, 131072 points per 16.0 seconds. Since we want, say, 32 seconds per stretch, we can either input 262144, or opt for 32.0. I never use the first option.

The program starts running. It first presents an overview of what will be done, which is also written in a log file called in this case event.log:

```
-----
Program gh_xte      Tue Jan  3 15:26:00 2012      User: belloni
-----
Working directory      :
/Users/belloni/data/xte/u1636/realtime/96087-01-84-10/pca
Input filename         :
/Users/belloni/data/xte/u1636/realtime/96087-01-84-10/pca/@event.lis
Instrument              : XTE/PCA
Observation ID         : 96087-01-84-10
Source name            : 4U_1636-53
Start date of observation : 2011-12-29T04:01:36

Channels for accumulation : 0-249
Type of output           : POWER
Output filename         : event.pds
Time resolution (s)      : 1/8192
Nyquist frequency (Hz)  : 4096.0000
Number of points per FFT : 262144
```

Walkthrough

The production program

```
Corresponding to length (s)      : 32.000000
Total number of FFTs             : 64
Time step                        : 32.000000
-----
```

Overview of the results

We have now produced a file called event.pds. This file contains the series of power density spectra as a function of time, the corresponding count rate for each of them (light curve), the photons from the spectral files in three energy bands (defaults here) and some housekeeping information.

The first thing we can do is have an overview of the content of the file. We can use the command

```
Ghats> ghats_all,'event.pds'
```

```
GH_info -----
```

```
PDS file name      : event.pds

Observatory        : XTE
Instrument          : PCA
Target             : 4U_1636-53

N. of trafos       : 64
Trafo duration     : 32.000000
N. of powers       : 262144
Start-end chans.   : 0-249
Starting mjd       : 55924.168
Time step          : 0.031250000
Barycentered       : NO
Spectral bins      : 3
Background         : NO
-----
```

The output includes some of the previous info, plus the last three lines which informs us that the input times were not barycentered, that the spectral bins used were three (default) and that there is no background associated to the PDS file. We'll see that later.

What is important is that our file contains a sequence of 64 PDS, each covering 32 seconds of uninterrupted data. You can obtain the same text output with the command

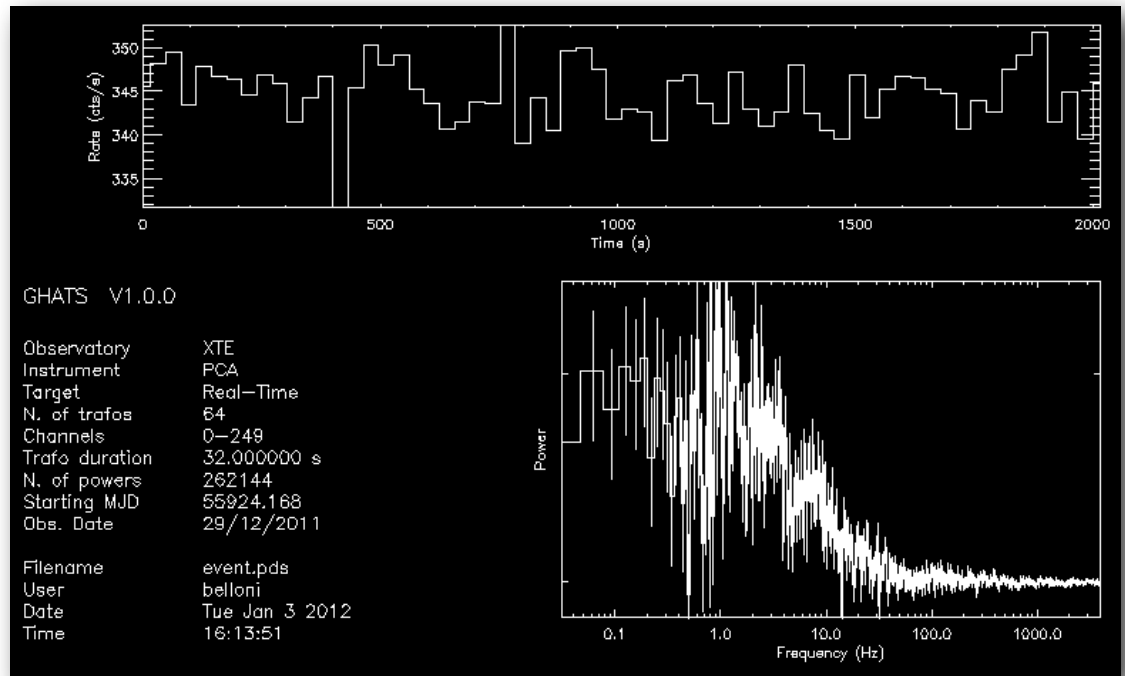
```
Ghats> gh_info,'event.pds'
```

Spaces between parts of the commands are harmless but not necessary. Here we add them for readability.

An important (and sorry) point for GDL users. If you use GDL, you can do away with typing the input file name, as for all analysis programs. If you use a variable which contains a non-existent filename or even which does not exist, a file selection window will be opened and the filename will be used and stored in that variable for further use. It is very convenient. Unfortunately, this does not work with GDL. The variable might be empty or contain a non-existent filename, but it must exist. To overcome this problem, I suggest you add to the ghats.pro routine (see startup command) a list of empty strings which you can use later.

Walkthrough

The `ghats_all` command, in addition to the text output, produces a summary plot which shows light curve (one count rate point per PDS) and the averaged PDS.



By default, the Poissonian level is not estimated and subtracted, the PDS rebinning is -100 (see below) and the output goes to the screen. You can have the Poissonian contribution subtracted with the keyword `/POISSON`, have a different rebinning factor with the keyword `REBIN` and produce a PostScript file (called `ghats.ps`) with the keyword `/PS`.

Light curve production

We are not happy with just looking at the data summary. We can extract the light curve and plot it with:

```
Ghats> gh_licu, 'event.pds', time, licu
```

```
Ghats> gh_plot_licu, time, licu
```

PDS extraction (with defaults)

Then we can extract the PDS ourselves, bin it and plot it with:

```
Ghats> ghx, 'event.pds', nu, pow, pow_e, /poisson
```

64 power spectra selected

This reads in the average PDS in three arrays: **nu** for frequency, **pow** for power and **pow_e** for error on the power. The **/poisson** keyword forces computation and subtraction of the Poissonian component.

```
Ghats> gh_reb, nu, pow, pow_e, -100, x, y, ye
```

This rebins the data with a rebin factor -100 and writes the output to the variables **x**, **y**, **ye**. Of course the choice of the variable names is arbitrary.

Finally, we can plot it:

```
Ghats> gh_plot_power, x, y, ye
```

Walkthrough

Custom PDS extraction

From the overview, we can see that the average count rate is around 340 cts/s, but we can use IDL/GDL to calculate it exactly from the light curve array we extracted:

```
Ghats> print, mean(licu)
344.632
```

Let us say that we are interested only in averaging the PDS corresponding to count rates larger than the average. We can do it with:

```
Ghats> ghx, 'event.pds', nu, pow, pow_e, /poisson, rate=[344.632, 10000]
33 power spectra selected
```

If we want to limit ourselves to the first 500 seconds of data, still selecting only count rates above the average:

```
Ghats> ghx, 'event.pds', nu, pow, pow_e, /poisson, rate=[344.632, 10000],
time=[0, 500]
10 power spectra selected
```

(Long commands may break to the page width in the Manual.)

We can also select only a range of indices:

```
Ghats> ghx, 'event.pds', nu, pow, pow_e, /poisson, index=[13, 34]
22 power spectra selected
```

Notice that GHATS commands assume indices start from 1 and not from 0.

Finally, we add yet another selection, only odd indices, making use of IDL/GDL:

```
Ghats> odd_indices = indgen(64) + findgen(32) + 1
Ghats> ghx, 'event.pds', nu, pow, pow_e, /poisson, rate=[344.632, 10000],
time=[0, 500], sel=odd_indices
7 power spectra selected
```

One more thing. The workhorse of GHATS is ghx. You can also extract an average PDS which is not Leahy normalized, but rms normalized. For this you need the count rate, which is available in the data file, and the background count rate in the same energy band, which you must estimate by other means. Once you have the background rate, say 13.76 counts/s, you can extract a rms normalized PDS with:

```
Ghats> Ghats> ghx, 'event.pds', nu, pow, pow_e, /poisson, rate=[344.632, 10000],
time=[0, 500], sel=odd_indices, rms=17.36
7 power spectra selected
```


Walkthrough - Cross spectrum

Cross spectra

In this part we show an example of creating the Real and Imaginary parts of the cross spectrum, plus the phase-lag spectrum and intrinsic coherence function, for light curves in two energy bands, as a function of Fourier frequency.

We start with an event file from *NICER*, that here we call **event.evt**. We first create the FFTs in the two energy bands, in this case 0.3-2 keV (PI 30-200) and 2.01-12 keV (PI 201-1200). Notice that we explicitly avoid that the bands overlap to avoid the partial correlation in the cross spectrum (below) due to the fact that some photons are in both bands:

Create FFTs

```
Ghats> gh_nicer, 'event.evt', 'FFT', [30,200], 10000,32768, 'evt_30-200.fft'  
Ghats> gh_nicer, 'event.evt', 'FFT', [201,1200], 10000,32768, 'evt_201-1200.fft'
```

Since the time resolution of *NICER* is 4×10^{-8} s, using a time rebin of 10000 yields a time resolution of 0.4 ms, which gives a Nyquist frequency of 1250 Hz. Using 32768, points at a resolution of 0.4 ms, for the FFT gives segments of 13.1072 s.

We now use **gh_cs** to compute the cross spectrum, the phase-lags and coherence:

```
Ghats> filer='evt_30-200.fft' & files='evt_201-1200.fft'
```

Here **filer** is the file with the FFT of the reference band (0.3-2 keV) and **files** is for the subject band (2.01-12 keV).

Create Real and Imaginary parts of the cross spectrum, the phase-lag spectrum and the coherence function

```
Ghats> gh_cs,filer, files, nu, ll, lle, cc, cce, -100, rr, rre, ii, iie, mw, qf,  
poisson=[400,800], rms1=0.001, rms2=0.001, /lowcoh  
67 FFTs selected  
Poisson level recomputed from average  
67 FFTs selected
```

% Program caused arithmetic error: Floating illegal operand

The error message in the last line can be safely ignored. There is no error in the procedure. This is likely due to the error routine of the coherence computing the sqrt of a negative number; this happens over a frequency range in which the coherence cannot be usefully computed, as explained below, so this is not important.

The above command returns the lags, array **ll**, coherence function, array **cc**, array real, array **rr**, and imaginary, array **ii**, parts of the cross spectrum with corresponding errors, arrays **lle**, **cce**, **rre** and **iie**, respectively, as a function of frequency (array **nu**). In this example, all the spectra are rebinned logarithmically (by a factor -100).

The array **mw** returns the product MW, where M is the number of trafos and W the number of frequencies rebinned.

Walkthrough

Quality factor flagging bad frequencies

The array **qf** returns the quality factor of each frequency bin; quality is considered good if the conditions described in Vaughan & Nowak (1997) are satisfied, and is set to bad if those conditions are not met (in the words of V&N97, the coherence cannot be usefully computed in those cases).

Renormalise cross spectrum to rms units

If the keywords **rms1** and **rms2** are given, the cross spectrum will be in rms units. If not given, the cross spectrum will be in Leahy units. If given, **rms1** and **rms2** should be the background rates in, respectively, the subject (1) and the reference (2) bands. In this example it was assumed that the backgrounds are insignificant (effectively 0; notice that you cannot put them to 0) compared to the source count rate.

Subtract Poisson level of the PDS and the average Real value from the Real vector over a given frequency range

The keyword **poisson** indicates the frequency range, in Hz, over which one computes the average Poisson level for the power spectra in the 2 bands and the average of the real part of the cross spectrum to subtract from the total real part of the cross spectrum. This last step corrects for two things: the effect of deadtime induced cross talk on the lags (see e.g., van der Klis et al. 1987 ApJL 319, L13) and the partial correlation introduced when the subject and reference bands overlap.

Make sure that the given range is within the limits of your FFT, but excludes the last bin (Nyquist frequency).

Compute errors of the coherence properly

The final keyword, **lowcoh**, computes the errors of the coherence using either case 1 (high power high observed coherence) or case 2 (high power low observed coherence) in V&N97. Case 3 (low power) is not implemented (also it is not available in V&N97), and frequency bins that fall in that case are flagged as bad (in the array **qf**). If this keyword is not given, the errors of the coherence are calculated using the case 1.

Finally, write everything in Xspec format.

Write Xspec files

```
Ghats> gh_xspec, freq, rr, rre, 'real_30-200_201-1200'  
Ghats> gh_xspec, freq, ii, iie, 'imag_30-200_201-1200'  
Ghats> gh_xspec, freq, ll, lle, 'plag_30-200_201-1200'  
Ghats> gh_xspec, freq, cc, cce, 'cohe_30-200_201-1200'
```

If we want to use the full band as reference band:

Use full band as reference band

```
Ghats> gh_nicer, 'event.evt', 'FFT', [30,1200], 10000, 32768, 'evt_30-1200.fft'  
Ghats> gh_nicer, 'event.evt', 'FFT', [200,500], 10000, 32768, 'evt_200-500.fft'
```

Walkthrough

```
Ghats> filer='evt_30-1200.fft & files='evt_200-500.fft
Ghats> gh_cs, filer, files, nu, ll, lle, cc, cce, -100, rr, rre, ii, iie, mw, qf,
poisson=[400,800], /lowcoh, /rotate
67 FFTs selected
Poisson level recomputed from average
67 FFTs selected
% Program caused arithmetic error: Floating illegal operand
```

Here we did not normalise the cross spectrum to rms units, so it is Leahy units, and added one more keyword, **rotate**. This rotates the cross vector by $\pi/4$ (45 degrees) counter clockwise. This may be useful if the real part is large and the imaginary part is small and potentially negative (the phase lags are small and potentially negative). This rotation will make the real and imaginary parts more or less equal, and the imaginary part will be positive so you can plot it in log scale in Xspec.

Notice that this only changes the real and imaginary parts, but the lag spectrum remains unrotated, and the coherence function is not affected.

If you apply this rotation and want to fit the real and imaginary parts in, e.g., Xspec, you will need to keep in mind this angle when predicting the lag spectrum. (The coherence is not affected.)

Example of fitting in Xspec

See an example of fitting the real and imaginary parts here: <https://github.com/figante/multi-lorentzian> (the example there is without rotation).

Internally, gh_cs uses gh_cross_re_im_new twice. Once to get the average value of the real part of the cross spectrum in the given range (using the keyword poisson=[freq_min,freq_max], as explained above), and the second time to subtract that average from the real part of the cross spectrum at all frequencies. If you want to have more control, you can use gh_cross_re_im_new yourself. For instance (assuming that **filer** and **files** are the same as above):

```
Ghats> gh_cross_re_im_new, filer, files, freq, lag, lag_err, coh, coh_err, -100, real,
real_err, imag, imag_err, rms1=0.001,rms2=0.001, /rotate
67 FFTs selected
% Program caused arithmetic error: Floating illegal operand
```

Then find the average of the Real part of the cross spectrum over a frequency range dominated by Poisson noise (in this case 400-800 Hz)

```
Ghats> poivalue=mean(real(where(freq ge 400 and freq le 800)))
```

And finally subtract that average real value from the Real part of the cross spectrum (this corrects both for cross talk and for the partial correlation if the subject band is part of the reference band):

Walkthrough

```
Ghats> gh_cross_re_im_new, filer, files, freq, lag, lag_err, coh, coh_err, -100, real,  
real_err, imag, imag_err, rms1=0.001, rms2=0.001, poivalue=poivalue, /rotate  
67 FFTs selected  
% Program caused arithmetic error: Floating illegal operand
```

Notice that, except for RXTE for which the Poisson level is calculated using the model of Zhang et al. (1995 ApJ 449 930), the program assumes that the Poisson level for the two power spectra in the calculation of the intrinsic coherence is 2.

To use the average power over some frequency range as the Poisson level, add the frequency range to average the power spectra, e.g.:

```
Ghats> gh_cross_re_im_new, filer, files, freq, lag, lag_err, coh, coh_err, -100, real,  
real_err, imag,imag_err, rms1=0.001, rms2=0.001, poivalue=poivalue,  
poisson=[400,800], /rotate  
67 FFTs selected  
% Program caused arithmetic error: Floating illegal operand
```

Since it does all this automatically for you, it is usually more convenient to use `gh_cs` instead of `gh_cross_re_im_new`.

List of GHATS commands

gh_xte	PDS/FFT extraction from RXTE/PCA files.
gh_swift	PDS/FFT extraction from Swift/XRT files.
gh_xmm	PDS/FFT extraction from XMM (pn & MOS) files.
gx_nustar	PDS/FFT extraction from NuSTAR files.
gh_nicer	PDS/FFT extraction from NICER files.
gh_laxpc_misra	PDS/FFT extraction from AstroSat/laxpc files.
gh_czti	PDS/FFT extraction from AstroSat/CZTI files.
gx_huiyan_le	PDS/FFT extraction from Huiyan (HXMT) LE files.
gx_huiyan_me	PDS/FFT extraction from Huiyan (HXMT) ME files.
gx_huiyan_he	PDS/FFT extraction from Huiyan (HXMT) HE files.
gahts/gghats	Startup procedure under IDL/GDL. Not for interactive use
gh_colors	Extract X-ray colors from a PDS file
gh_cross	Production of time/phase-lag spectrum and lag spectrum from two FFT
gh_cross_range	Extract phase/time lag around a feature
gh_dyn	Extraction of time-frequency image from PDS
gh_get_vle	Extract VLE flag from PCA HK data
gh_hk_xte	Extraction of housekeeping information from PDS/FFT for RXTE/PCA
gh_info	Output to terminal of basic PDS/FFT info
gh_licu	Extraction of light curve from PDS/FFT
gh_nspectra	Extraction of number of spectra for PDS/FFT
gh_oplot_nupower	Overplot of PDS in νP_ν form
gh_oplot_power	Overplot of PDS in P_ν form
gh pha	Production of PHA file from PDS
gh_plot_coh	Plot coherence
gh_plot_lag	Plot phase/time lags
gh_plot_hk_xte	Plot of summary of XTE HK information. Does not need previous gh_hk

List of GHATS commands

<code>gh_plot_licu</code>	Plot of light curve
<code>gh_plot_licu_nogaps</code>	Plot of light curve vs. array index, with gaps shown
<code>gh_plot_nupower</code>	Plot of PDS in νP_ν form
<code>gh_plot_power</code>	Plot of PDS in P_ν form
<code>gh_reb</code>	PDS rebinning
<code>gh_rmf</code>	production of (diagonal) response file to associate to PHA file for XSPEC
<code>gh_version</code>	Output of GHATS version
<code>gh_xspec</code>	Production of PHA/RSP file for XSPEC fitting. Incorporates <code>gh_ph</code> / <code>rmf</code>
<code>ghats_all</code>	Graphical overview of a PDS file
<code>ghats_getheader</code>	Extract PDS/FFT header. Not for interactive use
<code>ghats_openfft</code>	Open a FFT file. Not for interactive use
<code>ghats_openpds</code>	Open a PDS file. Not for interactive use
<code>ghrebin</code>	Low-level rebin procedure. Not for interactive use
<code>ghx</code>	Main PDS extraction procedure.
<code>poisson_estimate</code>	Calculate Poissonian contribution to PDS
<code>read_fft_line</code>	Read in a FFT from a FFT file
<code>read_pds_line</code>	Read in a PDS from a PDS file
<code>rebin_cross</code>	Rebin a lag spectrum
<code>gh_cros_re_im_new</code>	Computes lags, coherence, real and imaginary part of the cross spectrum
<code>gh_cs</code>	Uses <code>gh_cross_re_im_new</code> twice to correct Real part of cross spectrum, and to compute lags, coherence, real and imaginary part of the cross spectrum
<code>concatenate_pds</code>	Concatenates pds files (names given in an ASCII file) into a single pds file
<code>concatenate_fft</code>	Concatenates fft files (names given in an ASCII file) into a single fft file
<code>gh_dyn_fits</code>	Creates (and plots) the dynamical power spectrum (a.k.a. spectrogram)
<code>GH_dyn_fits_tk.py</code>	Python-GUI tool to plot dynamical power spectra (Spectrograms) created by <code>gh_dyn_fits</code> .

REFERENCE

GH_XTE

```
gh_xte, [/ghx], [/turbo],[bands=[s1,e1,s2,e2,s3,e3]],[gti=usergtifile], [/sliding], [/
    bary][,window=window][,wpar=wpar]
gh_xte,infilename,outtype,channels,treb,npds,oufilename, [/ghx], [/turbo],
    [bands=[s1,e1,s2,e2,s3,e3]],[gti=usergtifile], [/sliding], [/bary]
    [,window=window][,wpar=wpar]
gh_xte,parameterfile
```

Program for the production of PDS/FFT files starting from RXTE/PCA science/event array files. It is the main extraction procedure of the software.

infilename	Name of file in input. It can be an RXTE/PCA data file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,35])
treb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
oufilename	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
TURBO	If set, the program will use C. Markwardt's IDL_EXTRACT routine, running much faster. Not available under GDL.
BANDS	Optional channel bands for the production of three standard2 light curves. It is an array of six elements ([0,10,11,20,21,40]) means three curves in the PCA channel ranges 0-10, 11-20 and 21-40 respectively. They can then be used to produce X-ray hardnesses. Default values are [3,6,7,14,15,20].
GTI	optional user-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in RXTE time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_XTE (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_SWIFT

```
gh_swift, [/ghx], [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]  
    [,window=window] [,wpar=wpar]  
gh_swift, infilename, outtype, channels, reb, npds, outfile, [/ghx],  
    [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]  
    [,window=window] [,wpar=wpar]  
gh_swift, parameterfile
```

Program for the production of PDS/FFT files starting from Swift/XRT photon files. Both WT and PO files are accepted, although it is not possible to combine them. For Swift, the turbo flag is not available

infilename	Name of file in input. It can be a Swift/XRT photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,1023])
reb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
outfile	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the XRT channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [2,100,101,400,401,1023].
GTI	optional user-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in Swift time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_SWIFT (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_XMM

```
gh_xmm, [/ghx], gti=usergtifile, [bands=[s1,e1,s2,e2,s3,e3]], [/sliding], [/bary]  
    [,window=window][,wpar=wpar]  
gh_xmm,infilename,outtype,channels,treb,npds,oufilename, gti=usergtifile, [/ghx],  
    [bands=[s1,e1,s2,e2,s3,e3]], [/sliding], [/bary] [,window=window][,wpar=wpar]  
gh_xmm,parameterfile
```

Program for the production of PDS/FFT files starting from Swift/XRT photon files. Both WT and PO files are accepted, although it is not possible to combine them. For Swift, the turbo flag is not available

infilename	Name of file in input. It can be an XMM MOS or pn photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,1023])
treb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
oufilename	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [70,500,501,4000,4001,10000].
GTI	MANDATORY user-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in XMM time units), or a standard GTI FITS file. For XMM the GTI file is not optional, as it is the only way to obtain the GTI information.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_XMM (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_LAXPC_MISRA

```
gh_laxpc_misra, [/ghx], gti=usergtifile, [bands=[s1,e1,s2,e2,s3,e3]], [/sliding], [/bary] [,window=window] [,wpar=wpar]
gh_laxpc_misra,infilename,outtype,channels,treb,npds,oufilename, gti=usergtifile, [/ghx],[bands=[s1,e1,s2,e2,s3,e3]], [/sliding], [/bary] [,window=window] [,wpar=wpar]
gh_laxpc_misra,parameterfile
```

Program for the production of PDS/FFT files starting from Astrosat/LAXPC photon files. LAXPC units are treated as separate instruments. The string MISRA refers to the fact that the photon files must be one produced with the procedures by Ranjeev Misra.

infilename	Name of file in input. It can be a LAXPC photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,1023])
treb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
oufilename	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [16,43,44,113,114,257].
GTI	User-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in Astrosat time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_LAXPC_MISRA (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_CZTI

```
gh_czti, [/ghx], gti=usergtifile, [bands=[s1,e1,s2,e2,s3,e3]], [/sliding], [/bary]
        [,window=window][,wpar=wpar][,quadrants=quadrants][,fraction]
gh_czti,infilename,outtype,channels,treb,npds,oufilename, gti=usergtifile, [/ghx],
        [bands=[s1,e1,s2,e2,s3,e3]], [/sliding], [/bary] [,window=window][,wpar=wpar]
gh_czti,parameterfile
```

Program for the production of PDS/FFT files starting from Astrosat/CZTI photon files. CZTI units are treated as separate instruments.

infilename	Name of file in input. It can be a CZTI photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,500])
treb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
oufilename	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [20,80,81,256,257,500].
GTI	User-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in Astrosat time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_CZTI (cont'd)

QUADRANTS	array with the CZTI quadrants to be used for the accumulation. Default is [0,1,2,3]
FRACTION	value for the minimum exposure for CZTI pixels to be accepted. Default is 0.0
WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing; the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_NUSTAR

```
gh_nustar, [/ghx], [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]  
    [,window=window] [,wpar=wpar]  
gh_nustar,infilename,outtype,channels,treb,npds,oufilename, [/ghx],  
    [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]  
    [,window=window] [,wpar=wpar]  
gh_nustar,parameterfile
```

Program for the production of PDS/FFT files starting from NuSTAR photon files. The two telescopes can be combined through the use of metafiles.

infilename	Name of file in input. It can be a NuSTAR photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,1023])
treb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
oufilename	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [100,800,801,4095,100,4095].
GTI	optional user-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in NuSTAR time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_NUSTAR (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_NICER

```
gh_nicer, [/ghx], [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]  
        [,window=window][,wpar=wpar]  
gh_nicer,infilename,outtype,channels,treb,npds,oufilename, [/ghx],  
        [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]  
        [,window=window][,wpar=wpar]  
gh_nicer,parameterfile
```

Program for the production of PDS/FFT files starting from NICER photon files.

infilename	Name of file in input. It can be a NICER photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,1023])
treb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
oufilename	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [100,800,801,4095,100,4095].
GTI	optional user-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in NICER time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_NICER (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_HUIYAN_LE

```
gh__huiyan_le, [/ghx], [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/
    bary] [,window=window] [,wpar=wpar]
gh_ _huiyan_le, infilename, outtype, channels, reb, npds, outfile, [/ghx],
    [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]
    [,window=window] [,wpar=wpar]
gh_ _huiyan_le, parameterfile
```

Program for the production of PDS/FFT files starting from Huiyan/LE photon files.

infile	Name of file in input. It can be a Huiyan/LE photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,1023])
reb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
outfile	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [100,800,801,4095,100,4095].
GTI	optional user-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in HXMT time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_HUIYAN_LE (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_HUIYAN_ME

```
gh__huiyan_me, [/ghx], [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/
    bary] [,window=window] [,wpar=wpar]
gh_ _huiyan_me,infilename,outtype,channels,treb,npds,oufilename, [/ghx],
    [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]
    [,window=window] [,wpar=wpar]
gh_ _huiyan_me,parameterfile
```

Program for the production of PDS/FFT files starting from Huiyan/ME photon files.

infilename	Name of file in input. It can be a Huiyan/ME photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,1023])
treb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
oufilename	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [100,800,801,4095,100,4095].
GTI	optional user-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in HXMT time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_HUIYAN_ME (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GH_HUIYAN_HE

```
gh__huiyan_he, [/ghx], [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/
    bary] [,window=window] [,wpar=wpar]
gh_ _huiyan_he,infilename,outtype,channels,treb,npds,oufilename, [/ghx],
    [bands=[s1,e1,s2,e2,s3,e3]], [gti=usergtifile], [/sliding], [/bary]
    [,window=window] [,wpar=wpar]
gh_ _huiyan_he,parameterfile
```

Program for the production of PDS/FFT files starting from Huiyan/HE photon files.

infilename	Name of file in input. It can be a Huiyan/HE photon file, a metafile (starting with '@') or a metametafile (starting with '@@'). If the working directory is different from the data directory, full path names must appear in the (meta-)metafiles.
outtype	'POWER' for PDS production, 'FFT' for FFT production
channels	array with start-end values for PHA channels (example: [0,1023])
treb	Rebin factor in time (integer)
npds	If int/long: number of light curve points to be FFTed in each interval. If float/double: length of the intervals in seconds. It must be a power of two.
oufilename	Name of file in output. The extension should be 'pds' for PDS and 'fft' for FFT files
parameterfile	Name of parameter file. It should start with '#'. The ASCII file must contain the previous information, one per line
GHX	If set, ghats_all is launched at the end of the program.
BANDS	Optional channel bands for the production of three light curves for colors. It is an array of six elements ([0,100,101,200,201,400]) means three curves in the channel ranges 0-100, 101-200 and 201-400 respectively. They can then be used to produce X-ray hardnesses. Default values are [100,800,801,4095,100,4095].
GTI	optional user-supplied GTIs, to be intersected with the ones in the data files. If set, it must be a valid filename. Two possible input formats are possible. Either an ASCII file with lines containing start end time of each GTI (in HXMT time units), or a standard GTI FITS file.
SLIDING	optional parameter for sliding the time window. If not set, the time intervals for the FFT (data stretch) do not overlap. If set to N, they are shifted by an Nth of their total duration. As an example, if the data intervals are 16 seconds long, sliding =4 will shift two consecutive intervals by four seconds in start time. If this keyword is set, the averaged PDS obtained with GHX does not obviously make statistical sense.
BARY	flag for barycentered data. If set, the program will not read the TIME column, but BARYTIME. It is the user's responsibility to make sure that the BARYTIME column exists.

REFERENCE

GH_HUIYAN_HE (cont'd)

WINDOW	string with the name of the window function to apply to each stretch of data. Possible choices are currently: 'Boxcar' : rectangular window (no windowing: the default) 'Bartlett' 'Hann' 'Welch' 'Cosine' 'Hanning' 'Hamming' 'Triplet' 'Gauss' 'Kaiser' (see appendix A for a description of the functional shape of the windows)
WPAR	additional parameter for windows which require one (see appendix A)

REFERENCE

GHATS/GGHATS

```
ghats  
gghats
```

Program to start the package. It is usually not run interactively, but executed at startup of IDL/GDL

REFERENCE

GH_COLORS

```
gh_colors,filename,time,hr1,hr2[,rate1,rate2,rate3]
```

Program to extract X-ray color information from a PDS file.

filename	Input file name of PDS file
time	Output array with time values
hr1	Output array with HR1 values (rate2/rate1)
hr1	Output array with HR2 values (rate3/rate1)
rate1	Output light curve in band 1 (optional)
rate2	Output light curve in band 2 (optional)
rate3	Output light curve in band 3 (optional)

REFERENCE

GH_CROSS

(deprecated; use GH_ROSS_RE_IM_NEW and GH_CS below)

```
gh_cross,filename1,filename2,frequency,lag,lag_err,coh,coh_err,irf, [/tlag],  
[index=iarray], [time=tarray], [sel=selarray]
```

Program to read in two FFT from compatible FFT files, selecting on a range of FFTs in time, range, or selection of indices. It produces phase lags (in radians) or time lags, and coherence.

WARNING! In coherence computation, for RXTE/PCA data the Poissonian levels are estimated using precise expressions, while for other missions the Poissonian level is assumed to be 2.0.

Positive lags indicate that the second time series lags the first.

filename1	Name of file for FFT1.
filename2	Name of file for FFT2. It must be compatible with FFT1 (same observatory, instrument, target, time resolution, frequency resolution, start time and number of FFT)
frequency	output array containing the frequencies of the lag/coherence spectrum
lag	output array containing the phase/time lag spectrum
lag_e	output array containing the errors for the phase/time lag spectrum
coh	output array containing the coherence spectrum
coh_e	output array containing the errors for the coherence spectrum
irf	Rebinning factor (integer). If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/\text{irf})$. Typical negative value is -100.
TLAG	If set, the program will output time lags. If not, phase lags will be used.
INDEX	Optional index range for accumulated spectra. Example: index=[12,34] will average only spectra from #12 to #34 in the FFT file
TIME	Optional time range for accumulated spectra. Example: time=[0.0,500.0] will average only spectra corresponding to that time interval in the FFT file
SEL	Optional list of indices corresponding to spectra to accumulate. Example: sel=[1,3,5,7,9] will average only spectra corresponding to those indices in the FFT file. Remember that indices start from 1 in PDS/FFT files.

REFERENCE

GH_CROSS_RANGE

```
gh_cross_range,filename1,filename2,centroid,width,frequency,lag,lag_err,irf,  
[/tlag], [index=iarray], [time=tarray], [sel=selarray], [/plot],  
[/poilevel]
```

Program to read in two FFT from compatible FFT files, selecting on a range of FFTs in time, range, or selection of indices. It produces phase lags (in radians) or time lags averaged over a feature. For the feature, the user inputs centroid and FWHM and the average will be over the frequency range $[\text{centroid}-\text{FWHM}/2, \text{centroid}+\text{FWHM}/2]$. It optionally plots the full lag spectrum with the relevant interval marked.

WARNING! In coherence computation, for RXTE/PCA data the Poissonian levels are estimated using precise expressions, while for other missions the Poissonian level is assumed to be 2.0.

Positive lags indicate that the second time series lags the first.

filename1	Name of file for FFT1.
filename2	Name of file for FFT2. It must be compatible with FFT1 (same observatory, instrument, target, time resolution, frequency resolution, start time and number of FFT)
centroid	Centroid frequency for the averaging
width	FWHM for the averaging
frequency	output array containing the frequencies of the lag/coherence spectrum
lag	output array containing the phase/time lag spectrum
lag_e	output array containing the errors for the phase/time lag spectrum
irf	Rebinning factor (integer) for the lag plot. If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/\text{irf})$. Typical negative value is -100.
TLAG	If set, the program will output time lags. If not, phase lags will be used.
INDEX	Optional index range for accumulated spectra. Example: <code>index=[12,34]</code> will average only spectra from #12 to #34 in the FFT file
TIME	Optional time range for accumulated spectra. Example: <code>time=[0.0,500.0]</code> will average only spectra corresponding to that time interval in the FFT file
SEL	Optional list of indices corresponding to spectra to accumulate. Example: <code>sel=[1,3,5,7,9]</code> will average only spectra corresponding to those indices in the FFT file. Remember that indices start from 1 in PDS/FFT files.
PLOT	Optional keyword. If set, the full phase/time lag spectrum is plotted, rebinned according to irf, with vertical lines marking the averaging interval.
POILEVEL	Average value of the real part of the cross spectrum to be subtracted from the output value before computing the lag. It can be obtained with the same program giving in input a broad frequency range dominated by Poissonian noise

REFERENCE

GH_CROSS_RE_IM_NEW

```
gh_cross_re_im_new,filename1,filename2,frequency,lag,lag_err,coh,coh_err,irf, real,
    real_err, imaginary, imaginary_err, [rms1=r1], [rms2=r2], [/rotate],[/
    lowcoh],
    [/tlag], [index=iarray], [time=tarray], [sel=selarray], [poivalue=value],
    poisson=[nu1,nu2], [cross=cross]
```

Program to read in two FFT from compatible FFT files, selecting on a range of FFTs in time, range, or selection of indices. It produces phase lags (in radians) or time lags, and coherence.

WARNING! In coherence computation, for RXTE/PCA data the Poissonian levels are estimated using precise expressions, while for other missions the Poissonian level is assumed to be 2.0.

Positive lags indicate that the second time series lags the first.

filename1	Name of file for FFT1.
filename2	Name of file for FFT2. It must be compatible with FFT1 (same observatory, instrument, target, time resolution, frequency resolution, start time and number of FFT)
frequency	output array containing the frequencies of the lag/coherence spectrum
lag	output array containing the phase/time lag spectrum
lag_e	output array containing the errors for the phase/time lag spectrum
coh	output array containing the coherence spectrum
coh_e	output array containing the errors for the coherence spectrum
irf	Rebinning factor (integer). If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/irf)$. Typical negative value is -100.
real	Real part of the cross spectrum
real_err	Error of the real part of the cross spectrum
imaginary	Imaginary part of the cross spectrum
imaginary_err	Error of the imaginary part of the cross spectrum
RMS1	Optional value for the total average background rate in band 1 (filename1). When it is specified, the program will convert the powers from Leahy normalization to squared rms normalization.
RMS2	Optional value for the total average background rate in band 2 (filename2). When it is specified, the program will convert the powers from Leahy normalization to squared rms normalization. Notice that you have to give both RMS1 and RMS2; the program will stop if you give only one.
ROTATE	Optional keyword to rotate the cross vector by 45 degrees.

REFERENCE

GH_CROSS_RE_IM_NEW (cont'd)

LOWCOH	Optional keyword to use eq. 9 in Vaughan & Nowak 1997 in the case of high powers and low observed coherence (and eq. 8 in the case of high power and low observed coherence). If not given, the program computes errors always using eq. 8.
TLAG	If set, the program will output time lags. If not, phase lags will be used.
INDEX	Optional index range for accumulated spectra. Example: index=[12,34] will average only spectra from #12 to #34 in the FFT file
TIME	Optional time range for accumulated spectra. Example: time=[0.0,500.0] will average only spectra corresponding to that time interval in the FFT file
SEL	Optional list of indices corresponding to spectra to accumulate. Example: sel=[1,3,5,7,9] will average only spectra corresponding to those indices in the FFT file. Remember that indices start from 1 in PDS/FFT files.
POILEVEL	Average value of the real part of the cross spectrum to be subtracted from the output value before computing the lag. It can be obtained with the same program giving in input a broad frequency range dominated by Poissonian noise
POISSON	Frequency range, example poisson=[400,800], over which to compute the average Poisson level for the power spectra used in the calculation of the coherence.
CROSS	2-column COMPLEX array. Column 1: cross spectrum; column 2: errors on the real and imaginary parts of the cross spectrum.

REFERENCE

GH_CS

```
gh_cs,filename1,filename2,frequency,lag,lag_err,coh,coh_err,irf, real,real_err,  
    imaginary,imaginary_err,[rms1=r1],[rms2=r2],[/rotate],[/lowcoh],[/  
    tlag],[index=iarray],[time=tarray],[sel=selarray],[poivalue=value],  
    [cross=cross]
```

Program to read in two FFT from compatible FFT files, selecting on a range of FFTs in time, range, or selection of indices. It produces real and imaginary parts of the cross spectrum, phase lags (in radians) or time lags, and coherence.

This one is a wrapper that uses GH_CROSS_RE_IM_NEW twice. See walkthrough for explanation. The parameters are the same as those in GH_CROSS_RE_IM_NEW above.

REFERENCE

GH_DYN

```
gh_dyn,filename,time,rate,frequency,dynima, [index=iarray], [frebin=fr], [trebin=tr]
```

Program to read in a PDS file and extract a light curve and a time-frequency image

filename	Name of input PDS file.
time	Output array for time values
rate	Output array for count rate values
frequency	Output array containing the frequencies
dynima	Output array containing the time-frequency image
INDEX	Optional index range for accumulated spectra. Example: index=[12,34] will select and plot the dynamical PDS using only spectra from #12 to #34 in the PDS file
FREBIN	Frequency rebinning factor (integer). If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/\text{irf})$. Typical negative value is -100. Default is 1.
TREBIN	Time rebinning factor (integer). Only linear rebinning is allowed.

REFERENCE

GH_GETVLE

```
gh_getvle,filename,i_vle,[/nodialog]
```

Program to read in the RXTE/PCA VLE flag from a FH5a* file. Not intended for interactive use.

filename	Name of input PDS file.
----------	-------------------------

i_vle	Output VLE flag value
-------	-----------------------

NODIALOG	If this keyword is not set, no dialog window will be opened. Kind of reverse logic here.
----------	--

REFERENCE

GH_HK_XTE

```
gh_hk_xte,filename,time,vle,ndet,poisson
```

Program to read in a XTE/PCA PDS file and extract a housekeeping information for each spectrum, such as VLE rate, number of active detectors and estimated Poissonian level

filename	Name of input PDS file.
time	Output array for time values, one per PDS in the input file
vle	Output array with VLE (Very Large Events) rate values
ndet	Output array with number of active detectors
poisson	Output array with estimated Poisson value (using the expression from Zhang et al., 1995, ApJ, 449, 930.

REFERENCE

GH_INFO

```
gh_info, filename
```

Program to display basic information for a PDS/FFT file

filename	Name of input PDS or FFT file.
----------	--------------------------------

REFERENCE

GH_LICU

```
gh_licu,filename,time,rate, [delta=t], [mjd=rmjd0]
```

Program to read in a PDS file and extract a light curve

filename	Name of input PDS file.
time	Output array for time values
rate	Output array for count rate values
DELTA	Optional output keyword containing the time bin size for the light curve.
RMJD	Optional output keyword containing the MJD corresponding to time 0.

REFERENCE

GH_NSPECTRA

```
gh_nspectra,filename,n
```

Program to read in the number of PDS in a PDS file

filename	Name of input PDS file.
n	Output number of PDS

REFERENCE

GH_OPLOT_NUPOWER

```
gh_oplot_nupower, frequency, power, power_err
```

Program to overplot a PDS in νP_ν form

frequency	Array with input frequencies
power	Array with input powers
power_err	Array with input errors on powers

REFERENCE

GH_OPLOT_POWER

```
gh_oplot_power, frequency, power, power_err
```

Program to overplot a PDS in P_v form

frequency	Array with input frequencies
power	Array with input powers
power_err	Array with input errors on powers

REFERENCE

GH_PHA

```
gh_pha, power, power_err, phaname, rmfname, telescope, instrument
```

Program to write a PDS to a PHA file for XSPEC fitting. Not intended for interactive use (use GH_XSPEC)

power	Array with input powers
power_err	Array with input errors on powers
phaname	Filename for output PHA file
rmfname	Filename for associated RSP file, to be written into a keyword for automatic load together with the PHA
telescope	Input string with telescope name, for header (8 characters)
instrument	Input string with instrument name, for header (8 characters)

REFERENCE

GH_PLOT_COH

```
gh_plot_coh,frequency,coh,coh_err,[xlow,xhigh],[ylo,yhigh], [\ps]
```

Program to plot a light curve

frequency	Array with input frequencies
coh	Array with input coherence
coh_err	Array with input coherence errors
xlow	Optional low X value for plot (if specified, also xhigh must be given)
xhigh	Optional high X value for plot
ylo	Optional low Y value for plot (if specified, also yhigh must be given)
yhigh	Optional high Y value for plot
PS	Keyword for PostScript output. If set, the plot will go to a PS file called 'gh_coh.ps'

REFERENCE

GH_PLOT_LAG

```
gh_plot_lag,frequency,lag,lag_err,[xlow,xhigh],[ylow,yhigh], [/ps], [/time], [/lin], [/hyp]
```

Program to plot a light curve

frequency	Array with input frequencies
lag	Array with input phase lags
lag_err	Array with input phase-lag errors
xlow	Optional low X value for plot (if specified, also xhigh must be given)
xhigh	Optional high X value for plot
ylow	Optional low Y value for plot (if specified, also yhigh must be given)
yhigh	Optional high Y value for plot
PS	Keyword for PostScript output. If set, the plot will go to a PS file called 'gh_lag.ps'
TIME	Keyword: if set, input is time lags. The program changes only the Y label
LIN	Keyword: if set, plot lags on linear scale
HYP	Keyword: if set, the hyperbolic sine of the lags is plotted. This allows a log plot also for negative lags

REFERENCE

GH_PLOT_HK_XTE

```
gh_plot_hk_xte,filename, [\ps]
```

Program to plot XTE housekeeping information. It does not need gh_hk as it reads HK info directly from a PDS file

filename Input filename for PDS file

PS Keyword for PostScript output. If set, the plot will go to a PS file called 'gh_hk.ps'

REFERENCE

GH_PLOT_LICU

```
gh_plot_licu,time,rate,[xlow,xhigh],[ylow,yhigh], [\ps]
```

Program to plot a light curve

time	Array with input times
rate	Array with input count rates
xlow	Optional low X value for plot (if specified, also xhigh must be given)
xhigh	Optional high X value for plot
ylow	Optional low Y value for plot (if specified, also yhigh must be given)
yhigh	Optional high Y value for plot
PS	Keyword for PostScript output. If set, the plot will go to a PS file called 'gh_licu.ps'

REFERENCE

GH_PLOT_LICU_NOGAPS

```
gh_plot_licu_nogaps,time,rate,[xlow,xhigh],[ylow,yhigh], [\ps]
```

Program to plot a light curve where the time axis represents the index of the point in the array and the gaps are shown by vertical dotted lines

time	Array with input times
rate	Array with input count rates
xlow	Optional low X value for plot (if specified, also xhigh must be given)
xhigh	Optional high X value for plot
ylow	Optional low Y value for plot (if specified, also yhigh must be given)
yhigh	Optional high Y value for plot
PS	Keyword for PostScript output. If set, the plot will go to a PS file called 'gh_licu.ps'

REFERENCE

GH_PLOT_NUPOWER

```
gh_plot_nupower, frequency, power, power_err, [xlow, xhigh], [ylow, yhigh], [\ps]
```

Program to plot a PDS in vP_v form

frequency	Array with input frequencies
power	Array with input powers
power_err	Array with input errors on powers
xlow	Optional low X value for plot (if specified, also xhigh must be given)
xhigh	Optional high X value for plot
ylow	Optional low Y value for plot (if specified, also yhigh must be given)
yhigh	Optional high Y value for plot
PS	Keyword for PostScript output. If set, the plot will go to a PS file called 'gh_nupower.ps'

REFERENCE

GH_PLOT_POWER

```
gh_plot_power,frequency,power,power_err,[xlow,xhigh],[ylow,yhigh], [\ps]
```

Check syntax, change \ to /

Program to plot a PDS in P_v form

frequency	Array with input frequencies
power	Array with input powers
power_err	Array with input errors on powers
xlow	Optional low X value for plot (if specified, also xhigh must be given)
xhigh	Optional high X value for plot
ylow	Optional low Y value for plot (if specified, also yhigh must be given)
yhigh	Optional high Y value for plot
PS	Keyword for PostScript output. If set, the plot will go to a PS file called 'gh_power.ps'

REFERENCE

GH_REB

```
gh_reb, frequency, power, power_err, irf, nu, pow, pow_e
```

Program to rebin PDS linearly or logarithmically

frequency	Array with input frequencies
power	Array with input powers
power_err	Array with input errors on powers
irf	Frequency rebinning factor (integer). If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/\text{irf})$. Typical negative value is -100.
nu	Array with output rebinned frequencies
pow	Array with output rebinned powers
pow_e	Array with output errors on rebinned powers

REFERENCE

GH_RMF

```
gh_rmf, nulow, nuhigh, rmfname, telescope, instrument
```

Program to write a (diagonal) RSP file associated to a PHS file for XSPEC fitting. Not intended for interactive use (use GH_XSPEC)

nulow	Array with low boundaries for frequency bins
nuhigh	Array with high boundaries for frequency bins
rmfname	Filename for output RSP file.
telescope	Input string with telescope name, for header (8 characters)
instrument	Input string with instrument name, for header (8 characters)

REFERENCE

GH_VERSION

```
gh_version
```

Program to display in an external window the version number for GHATS.

REFERENCE

GH_XSPEC

```
gh_xspec, frequency, power, power_err, outputname, [\xspec], [telescope=tel],  
[instrument=inst]
```

Program to output a PDS in PHA form (with associated diagonal response in an RSP file) for XSPEC (or ISIS) fitting

frequency	Array with input frequencies
power	Array with input powers
power_err	Array with input errors on powers
outputname	File name for the output PHA and RSP files. No extension is required.
XSPEC	If set, this keyword will have XSPEC started automatically (if in the users's path) with the outputname file in input
TELESCOPE	Optional input telescope name, for header (8 characters)
INSTRUMENT	Optional input instrument name, for header (8 characters)

REFERENCE

GHATS_ALL

```
ghats_all,filename, [/ps], [/poisson], [rebin=reb]
```

Program to produce a summary plot from a PDS file. The plot includes text information, light curve and average power spectrum

filename	Input filename for PDS file
PS	Keyword for PostScript output. If set, the plot will go to a PS file called 'gh_power.ps'
POISSON	If set, the Poisson values computed with the formula of Zhang et al. (1995, ApJ, 449, 930) are subtracted
REBIN	Optional frequency rebinning factor (integer). If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/\text{irf})$. Default value if keyword is not set is -100.

REFERENCE

GHATS_GETHEADER

```
ghats_getheader,unit,gh_version_string,observatory,instrument,source,rmjd,nft,T,  
ntotal_ffts,e,sliding,baryflag,n_spectral_bins,background_flag,dummy
```

Program to extract header information from a PDS file. The file must have been already opened with ghats_openpds. Once opened, the file must be closed manually. Not intended for interactive use.

unit	Input file unit, received from ghats_openpds
gh_version_string	Output string with GHATS version information
observatory	Output string with observatory name
instrument	Output string with instrument name
source	Output string with source name
rmjd	Output MJD value for start time
nft	Output number of time bins per interval (twice the number of frequencies)
ntotal_ffts	Output number of PDS intervals contained in the file
e	Output 2*I array with start channel and end channel
sliding	Output value for the proliferation factor. In case sliding windows were chosen in gh_xte, this is the sliding factor used (see page for gh_xte).
baryflag	Output flag: 1=input times were barycentered
n_spectral_bins	Output number of spectral bins associated to each PDS. For RXTE this is fixed to three
background_flag	Output flag to inform whether additional background information is available in the file. For RXTE, this is always 0
dummy	Output array containing additional information.

REFERENCE

GHATS_OPENFFT

```
ghats_openfft,filename,unit, [/dialog]
```

Program to open an FFT file. It gives in output the unit associated to the file. The file must be close by the user. Not intended for interactive use.

filename	Input file name for FFT file
unit	Output file unit

DIALOG	If set, a dialog window for the file selection will be opened.
--------	--

REFERENCE

GHATS_OPENPDS

```
ghats_openpds,filename,unit, [/dialog]
```

Program to open a PDS file. It gives in output the unit associated to the file. The file must be close by the user. Not intended for interactive use.

filename	Input file name for PDS file
unit	Output file unit
DIALOG	If set, a dialog window for the file selection will be opened.

REFERENCE

GHREBIN

```
ghrebin,x,sx,y,sy,irf,xr,sxr,yr,syr,nrd
```

Program to rebin a PDS. Low level routine, not intended for interactive use. Use gh_reb to rebin.

x	Input array of frequency values
sx	Input array on frequency errors. If no errors are available, this variable is dummy (use x)
y	Input array of power values
sy	Input array on power errors. If no errors are available, this variable is dummy (use y)
irf	Input frequency rebinning factor (integer). If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/irf)$. Typical negative value is -100.
xr	Output array of rebinned frequency values
sxr	Output array of errors for xr
yr	Output array of rebinned power values
syr	Output array of errors for yr
nrd	Output number of elements of xr and yr

REFERENCE

GHX

```
ghx,filename,frequency,power,power_err, [index=in], [time=ti], [rate=ra], [sel=se],  
[/poisson], [rms=r]
```

Program to extract a PDS from a PDS file. The backbone of the analysis part of GHATS.

filename	Input file name of PDS file
frequency	Output frequency array
power	Output power array
power_err	Output array with errors on power
INDEX	Optional index range for accumulated spectra. Example: index=[12,34] will average only spectra from #12 to #34 in the FFT file
TIME	Optional time range for accumulated spectra. Example: time=[0.0,500.0] will average only spectra corresponding to that time interval in the FFT file
RATE	Optional rate range for accumulated spectra. Example: rate=[125.0,347.0] will average only spectra where the total count rate is between 125 and 347 cts/s.
SEL	Optional list of indices corresponding to spectra to accumulate. Example: sel=[1,3,5,7,9] will average only spectra corresponding to those indices in the FFT file. Remember that indices start from 0 in PDS/FFT files.
POISSON	If set, the Poisson values computed with the formula of Zhang et al. (1995, ApJ, 449, 930) are subtracted
RMS	Optional value for the total average background rate in the considered channel band. When it is specified, the program will convert the powers from Leahy normalization to squared rms normalization.

REFERENCE

POISSON_ESTIMATE

```
poisson_estimate,poisson_level,cnts,time_per_spectrum,current_vle_rate,nfreq,ndet,  
i_vle,tdead, [/differential]
```

Program to obtain the frequency-dependent RXTE/PCA Poissonian level after Zhang et al. (1995, ApJ, 449, 930). Not intended for interactive use.

poisson_level	Output array with frequency-dependent Poissonian level
cnts	Input count rate
time_per_spectrum	Input time length for the data interval
current_vle_rate	Input RXTE/PCA VLE rate
nfreq	Input number of frequencies in the PDS
ndet	Input number of active PCA detectors
i_vle	Input VLE flag
tdead	Input dead time value

DIFFERENTIAL If this keyword is set, the program returns the Poisson array for the current data stretch. If it is not set, the program adds the Poissonian array to the sum of the previous ones. The final division by the number of PDS in that case is left to the user.

REFERENCE

READ_FFT_LINE

```
read_fft_line,unit,muflag,rmjd,cnts,poisson,current_vle_rate,ndetectors,values
```

Program to read in the next FFT from an already opened FFT file. Not intended for interactive use.

unit	Input unit number for FFT file
muflag	not used
rmjd	Output MJD value corresponding to the start time of the current interval
cnts	Output count rate
poisson	Output Poissonian level (average value)
current_vle_rate	Output VLE rate
ndetectors	Output number of active detectors
values	Output array of FFT values (complex numbers)

REFERENCE

READ_PDS_LINE

```
read_pds_line,unit,muflag,rmjd,cnts,poisson,current_vle_rate,ndetectors,variance,values
```

Program to read in the next PDS from an already opened PDS file. Not intended for interactive use.

unit	Input unit number for PDS file
muflag	not used
rmjd	Output MJD value corresponding to the start time of the current interval
cnts	Output count rate
poisson	Output Poissonian level (average value)
current_vle_rate	Output VLE rate
ndetectors	Output number of active detectors
variance	Output zero-frequency power
values	Output array of PDS values

REFERENCE

REBINCROSS

```
rebincross,frequency,cross,irf,nu,rcross,nout
```

Program to rebin a cross spectrum linearly or logarithmically. Not intended for interactive use.

frequency	Array with input frequencies
cross	Array with cross powers
irf	Frequency rebinning factor (integer). If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/irf)$. Typical negative value is -100.
nu	Output array with output rebinned frequencies
rcross	Output array with output rebinned cross powers
nout	Output number of rebinned frequencies

REFERENCE

CONCATENATE_PDS

```
concatenate_pds, pdslst, outputfilename
```

Program to concatenate a number of PDS.

pdlist	Name of an ASCII file with the names of the PDS to concatenate, one per line. They should be in correct time order
outputfilename	Name of the file to save the concatenated PDS.

REFERENCE

CONCATENATE_FFT

```
concatenate_pds, fftlist, outputfilename
```

Program to concatenate a number of FFT files.

fftlist	Name of an ASCII file with the names of the FFT to concatenate, one per line. They should be in correct time order
outputfilename	Name of the file to save the concatenated FFT.

REFERENCE

GH_DYN_FITS

```
gh_dyn_fits, inputpds, outputfits, [index=iarray], [frebin=fr], [trebin=tr],  
[arguments], [/show]
```

Program to read in a PDS file, extract a light curve and a time-frequency image and save it to a FITS file.

If the user gives the option /show, this command will use *GH_DYN_FITS_TK.py*, the *GUI-python* interactive tool, to also plot the Spectrogram.

inputpds	Name of input PDS file.
outputfits	Output FITS file containing the time-frequency image
INDEX	Optional index range for accumulated spectra. Example: index=[12,34] will select only spectra from #12 to #34 in the PDS file
FREBIN	Frequency rebinning factor (integer). If positive, it will be used as a linear rebinning factor. If negative, the output will be rebinned logarithmically, i.e. each frequency bin will be larger than the previous one by a factor of $\exp(-1/\text{irf})$. Typical negative value is -100. Default is 1.
TREBIN	Time rebinning factor (integer). Only linear rebinning is allowed.
SHOW	If set, the <i>GUI-python</i> interactive tool <i>GH_DYN_FITS_TK.py</i> will be launched to plot the time-frequency image (a.k.a. Spectrogram). If not set, this command will write the file given in outputfits in FITS format and exit. In that case, you would have to plot the FITS file yourself, either using <i>GH_DYN_FITS_TK.py</i> separately, or any other plotting routine that works with FITS files.
ARGUMENTS	String of arguments that are passed to the <i>GUI-python</i> interactive tool <i>GH_DYN_FITS_TK.py</i> (must be written between quotes).

REFERENCE

GH_dyn_fits_tk.py

```
GH_dyn_fits_tk.py [-h] [-title TITLE] [-t0 TZERO] [-cmap CMAP] [-interp INTERPOLATOR]
                  [-pmin PMIN] [-pmax PMAX] [-fmin FMIN] [-fmax FMAX] [-ft TICK_FREQS
                  [TICK_FREQS ...]] [-fh HIGHLIGHT_FREQS [HIGHLIGHT_FREQS ...]] [-hc
                  HIGHLIGHT_COLOR] [-tmin TMIN] [-tmax TMAX] [-tt TICK_TIMES [TICK_TIMES ...]]
                  [-th HIGHLIGHT_TIMES [HIGHLIGHT_TIMES ...]] [-trebin TREBIN] [--savepdf] [--
                  savepng] [--noshow] [--dynPDSonly] [--nupnu] [--linear] [-cb] [--fillGaps]
                  [--showGaps] filename
```

GUI-python program that reads a FITS file containing a Spectrogram (a.k.a, dynamical PDS) created using *GH_DYN_FITS*, and opens an interactive plotting tool to plot and analyze the Spectrogram.

The program is available under the PYTHON directory, which must be in the PATH to be able to run.

filename	Name of input PDS file.
-h	Display a help message with all the available options.
-title	Choose a Window Title. Default is the Program Name and Version.
-t0	Set starting time. Default is 'TSTART' (or zero).
-cmap	Set the matplotlib colormap. Default 'viridis'.
-interp	Choose the matplotlib interpolator. Default 'gaussian'.
-pmin, pmax	Set minimum and maximum power (Leahy units). Can be set interactively afterwards.
-fmin, fmax	Set minimum and maximum frequency. Can be set interactively afterwards.
-ft	Set frequency ticks. Can be set interactively afterwards.
-fh	Set highlighted frequencies. Can be set interactively afterwards.
-tmin, tmax	Set minimum and maximum time. Can be set interactively afterwards.
-tt	Set time ticks. Can be set interactively afterwards.
-th	Set highlighted times. Can be set interactively afterwards.
-trebin	Set trebin value (for time label purposes only). Default is 1 or none.
--savepdf	Save the plot as a PDF file. Can also be saved using the save to PDF button.
--savepng	Save the plot as a PNG file. Can also be saved using the save to PNG button.
--noshow	Do not show the plot on the screen (useful for scripting purposes).
--dynPDSonly	Show only the Spectrogram, without the corresponding light curve.
--nupnu	Plot in nuPnu units instead of Pnu units.
--linear	Plot the Power in linear units.
--cb	Add the Power colorbar.
--fillGaps	Plot the full timespan include the GTI-gaps in the plot.
--showGaps	Plot compressed timespan but adding vertical lines at the GTI-gaps in the plot.

APPENDIX A

Window functions

Here are listed the functional forms of the available window functions. The functions here refer to the time interval $[-T/2, T/2]$ and are zero outside it. α , β , σ and γ are parameters which need to be supplied by the user.

Boxcar: $f(t) = 1$

Bartlett: $f(t) = 1 - \left| \frac{t - T/2}{T/2} \right|$

Hann: $f(t) = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi t}{T/2} \right) \right]$

Welch: $f(t) = 1 - \left(\frac{t - T/2}{T/2} \right)^2$

Cosine: $f(t) = \cos(\pi t)$

Hanning: $f(t) = \cos^2(\pi t)$

Hamming: $f(t) = \alpha + (1 - \alpha) \cos^2(\pi t)$

Triplet: $f(t) = e^{\frac{-\beta}{|t|}} \cos^2(\pi t)$

Gauss: $f(t) = \exp \frac{-1}{2} \frac{t^2}{\sigma^2}$

Kaiser: $f(t) = \gamma \sqrt{1 - (2t)^2}$